Hyperspectral Sparse Unmixing via Nonconvex Shrinkage Penalties

Longfei Ren, Danfeng Hong, *Senior Member, IEEE*, Lianru Gao, *Senior Member, IEEE*, Xu Sun, *Member, IEEE*, Min Huang, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Hyperspectral sparse unmixing aims at finding the optimal subset of spectral signatures in the given spectral library and estimating their proportions in each pixel. Recently, simultaneously sparse and low-rank representations (SSLRRs) have been widely used in the hyperspectral sparse unmixing task. This article developed a new unified framework to approximate the SSLRR-based unmixing model. The heart of the proposed framework is to design the new nonconvex penalties for efficient minimization by the means of two families of thresholding mappings, including the firm thresholding mapping and generalized shrinkage mapping. Both mappings can be regarded as generalizations of both soft thresholding and hard thresholding. Unlike previous approaches with explicit penalties, the proposed framework does not require explicit forms of penalties but only relevant thresholding mappings. Furthermore, an alternating direction method of multipliers (ADMM) was designed to solve the resulting optimization problem. Experiments conducted on the synthetic data and real data demonstrate the superiority of the proposed framework in improving the unmixing performance with respect to state-of-the-art methods.**

***Index Terms*— Alternating direction method of multipliers (ADMM), hyperspectral image, nonconvex optimization, sparse unmixing, thresholding mapping.**

I. Introduction

HYPERSPECTRAL images are acquired using spaceborne or airborne sensors that record reflected light in hundreds of narrowbands [1]. These bands contain the visible light spectrum, which includes short-wave infrared, long-wave infrared, and ultraviolet. The benefit of receiving spectral reflectance values over many bands is that it encapsulates more spectral information about the different materials present in the hyperspectral images [2], [3]. This has led to a growing interest in hyperspectral data analysis, such as classification [4], [5], [6], [7], denoising [8], [9], [10], [11], object/anomaly detection [12], [13], [14], superresolution [15], [16], and spectral unmixing [17], [18], [19], [20].

The extra spectral information leads to a tradeoff of significantly reduced spatial information, resulting in mixed pixels often appearing in hyperspectral images. To some degree, the existence of mixed pixels restricts the application of hyperspectral images. Spectral unmixing is a crucial technique in hyperspectral image exploitation, which aims at extracting the pure spectral signatures (or endmembers) and estimating their proportions (or abundances) in each pixel [21]. The extensive majority of unmixing methods consider the linear mixing model (LMM), which assumes that the influences of secondary reflection and multiple scattering on the materials are ignored, and the spectral responses of the distinct signature bear no interference with each other.

Based on the LMM, spectral unmixing methods can be classified as supervised, unsupervised, and semisupervised methods in accordance with the prior information of the endmember. The supervised-based unmixing methods require the endmembers in the images to be completely given. Then, these methods are essentially equivalent to the abundance estimation methods, which are usually solved by the least-squares algorithm. For the unsupervised-based unmixing methods, both the endmember information and the abundance information are unknown. Many unsupervised approaches based on geometry [22], [23], [24], statistic [25], [26], and nonnegative matrix factorization (NMF) [27], [28] have been developed and achieved great progress. Nevertheless, such approaches may encounter obstacles when the pixels are heavily mixed or the pure pixels do not exist in hyperspectral images. Also, such approaches may obtain virtual endmembers with no physical meaning. The semisupervised-based unmixing methods also known as sparse representation (SR)-based methods, on the other hand, have emerged as a compromise between supervised and unsupervised methods. The core assumption of the semisupervised-based unmixing methods is that the mixed pixels are represented by linear combinations of a small number of endmembers from a spectral library that is known in advance. Unmixing then is equivalent to picking the best subset of signatures in the spectral library.

Many approaches based on the SR technique have been proposed in the literature. The sparse unmixing via variable

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splitting and augmented Lagrangian (SUnSAL) [29] and collaborative SUnSAL (CLSUnSAL) [30] are the most common convex models, which constrain the single sparsity or the global row sparsity on the abundances by means of the $l_{1,1}$ and $l_{2,1}$ norms, respectively. A number of works further exploit the sparsity of the abundances by introducing the weighting factors [31], [32], [33], [34] or other nonconvex regularizers, such as the $l_q(0 \leq q < 1)$ regularizer [35], [36] to ameliorate the unmixing performance. From the spectral variability point of view, several works incorporating the group structure of the endmember in the spectral library have been developed [37], [38]. To account for the spatial information, two regularizers have been widely embedded in the unmixing model. On the one hand, the total variation (TV) regularizer [39], [40] and its variants [41] have been developed to promote spatial homogeneity and local smoothness between adjacent pixels. However, the TV regularizer requires that adjacent pixels should have similarly mixed endmembers and abundances, which is quite restrictive. The nonconvex low-rank regularizer, on the other hand, is another stimulating manner to exploit spatial information. It has been increasingly adopted in image processing [10], [11], [14] and the unmixing model [42], [43], [44], [45]. The hyperspectral unmixing by reweighted low rank and TV (HURLR-TV) [43] model combines the TV and weighted nuclear norm together for the unmixing problem. Recent work studying TV in deblurring and sparse unmixing of hyperspectral images opened a new perspective and can be important in future work [46]. The alternating direction sparse and low-rank unmixing (ADSpLRU) [44] model simultaneously imposes single sparsity and low rankness on the abundances adopting the weighted l_1 and nuclear norms. The joint-sparse-blocks and low-rank unmixing (JSpBLRU) [45] model adopts both the weighted $l_{2,1}$ and nuclear norms to further exploit the spatial information in hyperspectral imagery. Despite the good performance achieved with simultaneously sparse and low-rank representation (SSLRR)-based unmixing approaches, most methods directly substitute the low-rank regularizer with weighted nuclear norm resulting in insufficient expression for the spatial correlation.

Recent work in [38] has proposed a new manner to enhance the sparsity of solutions for the unmixing model. A nonconvex penalty function, which can be regarded as a close approximation of the l_0 norm, was designed for efficient minimization by means of the generalized shrinkage mapping. Motivated by the excellent results in [38], we propose a new unified framework called hyperspectral sparse unmixing via nonconvex shrinkage penalties (HSUNSP) in this article. The proposed framework contains four different manners to approximate the unmixing model. Two families of thresholding mappings, including the firm thresholding mapping [47], [48] and generalized shrinkage mapping [49], which are the closed-form proximal mappings of the nonconvex shrinkage-induced penalties, have been embedded in our framework. Both thresholding mappings can be regarded as generalizations of both soft thresholding and hard thresholding. An important aspect of the unified framework is that it does not need an explicit form of the penalties but only the associated thresholding mappings. More specifically, the main contributions of this article can be summarized as follows.

1) We proposed a unified framework HSUNSP utilizing the firm thresholding mapping or generalized shrinkage mapping to approximate the SSLRR-based unmixing model. The proposed framework consists of four different versions: two versions are the ameliorations of ADSpLRU, and another two are the ameliorations of JSpBLRU.
2) To the best of our knowledge, this is the first time to employ the nonconvex shrinkage-induced penalties as the approximations for the low-rank regularizer instead of the weighted nuclear norm in the unmixing model. The derived framework provides strong applicability and flexibility for most state-of-the-art unmixing algorithms than having explicit penalties.
3) We design an alternating direction method of multipliers (ADMM) to optimize the proposed model, remarkably improving unmixing performance without sacrificing computational efficiency.

The rest of this article is organized as follows. Section II reviews the related work. Section III describes the proposed framework. Section IV shows experimental results obtained on both synthetic data and real data. The conclusion is drawn in Section V. The Appendix gives the proof of Theorem 1.

II. Related Work

In this section, we review the LMM and the related state-of-the-art unmixing models, including the SR- and SSLRR-based models.

A. LMM

The LMM is a basic model to handle unmixing problems, which assumes that the mixed pixel $\mathbf{y}_i$ can be expressed as a linear combination of several endmembers weighted by their abundances [21]

$$\mathbf{y}_i = \mathbf{M}\mathbf{s}_i + \mathbf{n}_i \tag{1}$$

where $\mathbf{y}_i := [\mathbf{y}_{i,1}, \mathbf{y}_{i,2}, \ldots, \mathbf{y}_{i,L}]^T \in \mathbb{R}^{L\times 1}$ denotes the L-spectrum measured pixel; $\mathbf{M} := [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_e] \in \mathbb{R}^{L\times e}$ denotes the endmember matrix containing e endmembers; each $\mathbf{m}_j := [\mathbf{m}_{j,1}, \mathbf{m}_{j,2}, \ldots, \mathbf{m}_{j,L}]^T \in \mathbb{R}^{L\times 1}$ $(j = 1, 2, \ldots, e)$ denotes the L-spectrum of the jth endmember; $\mathbf{s}_i := [\mathbf{s}_{i,1}, \mathbf{s}_{i,2}, \ldots, \mathbf{s}_{i,e}]^T \in \mathbb{R}^{e\times 1}$ denotes the abundances associated with the endmember matrix; and $\mathbf{n}_i := [\mathbf{n}_{i,1}, \mathbf{n}_{i,2}, \ldots, \mathbf{n}_{i,L}] \in \mathbb{R}^{L\times 1}$ denotes the noise vector.

The SR-based approaches replace the endmember matrix $\mathbf{M}$ with a known spectral library $\mathbf{A}$. Unmixing then amounts to finding the optimal subset in the library. Especially, the linear sparse mixing model can be represented as [29]

$$\mathbf{y}_i = \mathbf{A}\mathbf{x}_i + \mathbf{n}_i \tag{2}$$

where $\mathbf{x}_i \in \mathbb{R}^{m\times 1}$ represents the abundances associated with the spectral library $\mathbf{A} \in \mathbb{R}^{L\times m}$. Let $\mathbf{Y} := [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n] \in \mathbb{R}^{L\times n}$, $\mathbf{X} := [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \mathbb{R}^{m\times n}$, and $\mathbf{N} := [\mathbf{n}_1, \mathbf{n}_2, \ldots, \mathbf{n}_n] \in \mathbb{R}^{L\times n}$ denote the whole observed hyperspectral image, the

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corresponding abundances matrix, and the noise matrix; the matrix form of (2) can be formulated as

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{N}. \tag{3}$$

B. SR-Based Models

As the number of endmembers in the real scenarios is far less than that in the library, i.e., the abundances corresponding to the library are sparse, the SUnSAL model adopts the $l_{1,1}$ norm as a sparse regularizer [29]

$$\begin{aligned} \min_{\mathbf{X}} \quad & \frac{1}{2}\|\mathbf{A}\mathbf{X} - \mathbf{Y}\|_F^2 + \lambda\|\mathbf{X}\|_{1,1} \\ \text{s.t.} \quad & \mathbf{X} \geq 0 \end{aligned} \tag{4}$$

where $\|\cdot\|_F$ represents the Frobenius norm; $\|\mathbf{X}\|_{1,1} := \sum_{j=1}^{n}\sum_{i=1}^{m}|\mathbf{X}_{i,j}|$ with $\mathbf{X}_{i,j}$ being an element in the ith row jth column of abundances matrix $\mathbf{X}$; and λ represents the regularization parameter.

Assuming that all pixels in given hyperspectral imagery share the same active set of endmembers, i.e., present a similar sparsity pattern, the CLSUnSAL model constrains the global row sparsity to all pixels by means of the $l_{2,1}$ norm, that is [30]

$$\begin{aligned} \min_{\mathbf{X}} \quad & \frac{1}{2}\|\mathbf{A}\mathbf{X} - \mathbf{Y}\|_F^2 + \lambda\|\mathbf{X}\|_{2,1} \\ \text{s.t.} \quad & \mathbf{X} \geq 0 \end{aligned} \tag{5}$$

where $\|\mathbf{X}\|_{2,1} := \sum_{i=1}^{m}\|\mathbf{X}(i,:)\|_2$ with $\mathbf{X}(i,:)$ being the ith row of abundances matrix $\mathbf{X}$.

C. SSLRR-Based Models

By considering that all pixels in a small sliding window are correlated, i.e., the column of the abundances has a linear correlation, the low-rank representation has been developed to account for the spatial correlation. The ADSpLRU model adopts the weighted $l_{1,1}$ and nuclear norms to impose both single sparsity and low rankness on the abundances [44]

$$\begin{aligned} \min_{\mathbf{X}\in\mathbb{R}^{m\times K}} \quad & \frac{1}{2}\|\mathbf{A}\mathbf{X} - \mathbf{Y}\|_F^2 + \lambda\|\mathbf{W}\circ\mathbf{X}\|_{1,1} + \tau\|\mathbf{X}\|_{\mathbf{b},*} \\ \text{s.t.} \quad & \mathbf{X} \geq 0 \end{aligned} \tag{6}$$

where K denotes the number of pixels contained in a small sliding window; $\circ$ denotes the Hadamard product; λ and τ denote the regularization parameters; $\|\mathbf{W}\circ\mathbf{X}\|_{1,1} := \sum_{i=1}^{m}\sum_{j=1}^{K}\mathbf{W}_{i,j}|\mathbf{X}_{i,j}|$ with $\mathbf{W}_{i,j}$ being an element in the ith row jth column of the nonnegative weighting matrix $\mathbf{W}$; and $\|\mathbf{X}\|_{\mathbf{b},*} := \sum_{i=1}^{\operatorname{rank}(\mathbf{X})}\mathbf{b}_i\sigma_i(\mathbf{X})$ with $\mathbf{b} := [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_r]$ being the nonnegative weighting vector and $\sigma_i(\mathbf{X})$ being the ith singular value of $\mathbf{X}$.

To further utilize the spatial information in hyperspectral imagery, the JSpBLRU model adopts the weighted $l_{2,1}$ and nuclear norms to exploit the joint-sparsity-blocks' structure and low rankness structure of abundances [45]

$$\begin{aligned} \min_{\mathbf{X}\in\mathbb{R}^{m\times K}} \quad & \frac{1}{2}\|\mathbf{A}\mathbf{X} - \mathbf{Y}\|_F^2 + \lambda\sum_{j=1}^{s}\left\|\mathbf{X}_j\right\|_{\mathbf{w}_j,2,1} + \tau\|\mathbf{X}\|_{\mathbf{b},*} \\ \text{s.t.} \quad & \mathbf{X} \geq 0 \end{aligned} \tag{7}$$

where $\mathbf{X} := [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_s]$ can be partitioned into $s(1 \leq s \leq K)$ group, and $\mathbf{X}_j \in \mathbb{R}^{m\times d_j}(j = 1, 2, \ldots, s)$, $\sum_{j=1}^{s} d_j = K$; $\|\mathbf{X}_j\|_{\mathbf{w}_j,2,1} := \sum_{i=1}^{m}\mathbf{w}_{i,j}\|\mathbf{X}_j(i,:)\|_2$ with $\mathbf{w}_j := [\mathbf{w}_{1,j}, \mathbf{w}_{2,j}, \ldots, \mathbf{w}_{m,j}]$ being the nonnegative weighting vector of the joint-sparsity-blocks' structure.

Both SR- and SSLRR-based approaches can be regarded as the approximation of the following model:

$$\begin{aligned} \min_{\mathbf{X}} \quad & \frac{1}{2}\|\mathbf{A}\mathbf{X} - \mathbf{Y}\|_F^2 + \lambda\|\mathbf{X}\|_c + \tau\ \operatorname{rank}(\mathbf{X}) \\ \text{s.t.} \quad & \mathbf{X} \geq 0 \end{aligned} \tag{8}$$

where $c = 0$ or $\{2, 0\}$; $\|\mathbf{X}\|_0$ and $\|\mathbf{X}\|_{2,0}$ count the number of nonzero elements or the number of nonzero rows in $\mathbf{X}$. The problem (8) is NP-hard, and therefore, it is arduous to solve in a straightforward way. One common way is to approximate problem (8) using convex or easy-to-solve nonconvex penalties. The $l_{1,1}$ norm in the SU model and the weighted $l_{1,1}$ norm in the ADSpLRU model are approximations to $\|\cdot\|_0$; the $l_{2,1}$ norm in CLSUnSAL model and the weighted $l_{2,1}$ norm in JSpBLSU model are approximations to $\|\cdot\|_{2,0}$; and the weighted nuclear norm in ADSpLRU and JSpBLSU models is an approximation to $\operatorname{rank}(\cdot)$.

III. Proposed Framework

In this section, we first introduce the nonconvex shrinkage-induced penalties that are the key ingredients in our proposed framework. Then, we derive the new approximations of (8).

A. Nonconvex Shrinkage-Induced Penalties

As described before, nonconvex regularizers have shown significant potential in unmixing performance than convex ones. Herein, we design several nonconvex shrinkage-induced penalties to approximate the unmixing model. In order to efficiently minimize the nonconvex shrinkage-induced penalties, we introduce two families of thresholding mappings, which are the closed-form proximal mappings of the nonconvex penalties. The first thresholding mapping that we consider is the firm thresholding mapping, which first appeared in [47] to connect the WaveShrink procedure for denoising and nonparametric regression.

Definition 1: For a given matrix $\mathbf{C} \in \mathbb{R}^{m\times n}$, $\lambda > 0$, and $\mu > \lambda$, the firm thresholding mapping $S_{\text{firm},\lambda,\mu}$ from $\mathbb{R}^{m\times n}$ to $\mathbb{R}^{m\times n}$ is defined by

$$S_{\text{firm},\lambda,\mu}(\mathbf{C}) := \operatorname{sign}(\mathbf{C}) \circ s_{\text{firm},\lambda,\mu}(|\mathbf{C}|) \tag{9}$$

where

$$s_{\text{firm},\lambda,\mu}(\mathbf{C})_{i,j} := \begin{cases} 0, & \text{if } \mathbf{C}_{ij} \leq \lambda \\ \dfrac{\mu}{\mu-\lambda}\left(\mathbf{C}_{ij} - \lambda\right), & \text{if } \lambda \leq \mathbf{C}_{ij} \leq \mu \\ \mathbf{C}_{ij}, & \text{if } \mathbf{C}_{ij} \geq \mu \end{cases} \tag{10}$$

is the firm function.

Other kinds of thresholding mapping that we consider are generalized shrinkage mappings, which first appeared in [49] to reconstruct MRI from very little data.

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Definition 2: For a given matrix $\mathbf{C} \in \mathbb{R}^{m\times n}$ and $\lambda > 0$, the shrinkage mappings $S_{a,\lambda}$ and $S^g_{a,\lambda}$ from $\mathbb{R}^{m\times n}$ to $\mathbb{R}^{m\times n}$ are defined by

$$S_{a,\lambda}(\mathbf{C}) := \text{sign}(\mathbf{C}) \circ s_{a,\lambda}(|\mathbf{C}|) \tag{11}$$

$$S^g_{a,\lambda}(\mathbf{C})(i,:) := \frac{\mathbf{C}(i,:)}{\|\mathbf{C}(i,:)\|_2} s^g_{a,\lambda}(\mathbf{C}(i,:)) \tag{12}$$

where $a \leq 1$, $s_{a,\lambda}(\mathbf{C}) := \max(\mathbf{C} - \lambda^{2-a}\mathbf{C}^{a-1}, 0)$, and $s^g_{a,\lambda}(\mathbf{C}(i,:)) := \max(\|\mathbf{C}(i,:)\|_2 - \lambda^{2-a}\|\mathbf{C}(i,:)\|_2^{a-1}, 0)$ $(i = 1, 2, \ldots, m)$ are the shrinkage functions.

Our motivation for considering the thresholding mapping is to have them as explicit proximal mappings of the nonconvex shrinkage-induced penalties. The following propositions presented in [50] and [51] record some properties of the nonconvex shrinkage penalty functions induced by the thresholding mappings.

Proposition 1: Suppose that $s : [0, \infty) \to \mathbb{R}$ is continuous and satisfies $c \leq \lambda$; then, $s(c) = 0$ for some $\lambda \geq 0$ is strictly increasing on $[\lambda, \infty)$ and $s(c) \leq c$. Define $S(\mathbf{C}) := \text{sign}(\mathbf{C}) \circ s(|\mathbf{C}|)$. Then, there exists a real-valued function G such that

$$\underset{\mathbf{X}}{\text{argmin}} \frac{1}{2}\|\mathbf{X} - \mathbf{C}\|_F^2 + \lambda G(\mathbf{X}) = S(\mathbf{C}) \tag{13}$$

where $G(\mathbf{X}) = \sum_{i,j} g(\mathbf{X}_{ij})$ and g is even, strictly increasing, and continuous on $[0, \infty)$, differentiable except at 0 with $\partial g(0) = [-1, 1]$ if and only if $\lambda > 0$. If $c - s(c)$ is nonincreasing on $[\lambda, \infty)$, then g is concave on $[0, \infty)$, and G satisfies the triangle inequality; $S(\cdot)$ is the proximal mapping of $G(\cdot)$.

Proposition 2: Let $\lambda > 0$, $a \in \mathbb{R}$, and define g as in Proposition 1. Then,

$$\underset{\mathbf{X}}{\text{argmin}} \frac{1}{2}\|\mathbf{X} - \mathbf{C}\|_F^2 + \lambda g\big(\|\mathbf{X}\|_{2,1}\big) = S^g_{a,\lambda}(\mathbf{C}) \tag{14}$$

where $S^g_{a,\lambda}(\cdot)$ is the proximal mapping of $g(\cdot)$.

Note that both the firm thresholding and generalized shrinkage mappings satisfy all hypotheses of Proposition 1. The penalty function G_f induced by the firm thresholding mapping has a closed-form [50], while the penalty function G_s induced by the generalized shrinkage mapping cannot be written explicitly except for a few special values of a. In fact, we do not need to calculate the penalty function exactly but only the explicit proximal mapping of the penalty function, and we can do so efficiently using the firm thresholding or generalized shrinkage mappings. The derived framework based on the nonconvex shrinkage penalties is more flexible for most state-of-the-art algorithms for unmixing than having explicit penalties.

The firm thresholding mapping is a piecewise linear approximation of the hard thresholding mapping. For a given matrix $\mathbf{C} \in \mathbb{R}^{m\times n}$ and $\lambda > 0$, the hard thresholding mapping $H_\lambda(C)$ from $\mathbb{R}^{m\times n}$ to $\mathbb{R}^{m\times n}$ is defined by

$$H_\lambda(C) := \begin{cases} 0, & \text{if } |\mathbf{C}_{ij}| \leq \lambda \\ \mathbf{C}_{ij}, & \text{if } |\mathbf{C}_{ij}| > \lambda. \end{cases} \tag{15}$$

The hard thresholding mapping is well known as the proximal mapping of l_0 norm [50]

$$\begin{aligned} \text{Prox}_{\lambda\|\cdot\|_0} &:= \underset{\mathbf{X}}{\text{argmin}} \frac{1}{2}\|\mathbf{X} - \mathbf{C}\|_F^2 + \lambda\|\mathbf{X}\|_0 \\ &= H_{\sqrt{2\lambda}}(\mathbf{C}). \end{aligned} \tag{16}$$

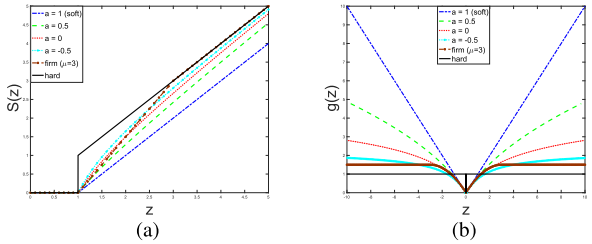


Fig. 1. (a) Plot of several thresholding mappings ($\lambda = 1$). (b) Plot of several penalty functions induced by several thresholding mappings ($\lambda = 1$).

When $a = 1$, the generalized shrinkage mappings $S_{a,\lambda}$ and $S^g_{a,\lambda}$ are the proximal mappings of $l_{1,1}$ (also known as soft thresholding) and $l_{2,1}$ norm [52], [53]

$$\begin{aligned} \text{Prox}_{\lambda\|\cdot\|_{1,1}} &:= \underset{\mathbf{X}}{\text{argmin}} \frac{1}{2}\|\mathbf{X} - \mathbf{C}\|_F^2 + \lambda\|\mathbf{X}\|_{1,1} \\ &= S_{1,\lambda}(\mathbf{C}) \end{aligned} \tag{17}$$

$$\begin{aligned} \text{Prox}_{\lambda\|\cdot\|_{2,1}} &:= \underset{\mathbf{X}}{\text{argmin}} \frac{1}{2}\|\mathbf{X} - \mathbf{C}\|_F^2 + \lambda\|\mathbf{X}\|_{2,1} \\ &= S^g_{1,\lambda}(\mathbf{C}). \end{aligned} \tag{18}$$

Note that $\lim_{\mu\to\infty} S_{\text{firm},\lambda,\mu}(\mathbf{C}) = S_{1,\lambda}(\mathbf{C})$ pointwise, and $\lim_{\lambda\to-\infty} S_{a,\lambda}(\mathbf{C}) = H_\lambda(\mathbf{C})$ pointwise; thus, both firm thresholding mapping $S_{\text{firm},\lambda,\mu}$ and generalized shrinkage mapping $S_{a,\lambda}$ can be regarded as generalizations of both soft thresholding and hard thresholding.

To further depict the relationship between the thresholding mappings and, hence, the induced penalties, Fig. 1 plots the numerically computed examples. As shown in Fig. 1(a), the firm thresholding removes the bias for large inputs without the discontinuity of hard thresholding. The smaller a in generalized shrinkage mappings, the smaller the bias imposes on large inputs. The hard thresholding applies no bias on large inputs; nevertheless, its discontinuity makes it quite unstable when used with most state-of-the-art algorithms [54]. Fig. 1(b) shows the penalty functions induced by several thresholding mappings with λ equal to 1. As one can see, the firm thresholding-induced penalty function is quadratic near the origin and then constant. The smaller the value of a, the slower the growth of the generalized shrinkage-induced penalty functions, being bounded above when $a < 0$. For $a \neq 0$, g approaches $|z|^a/a + c$ for large $|z|$ for some constant c (or $\log|z| + c$ for $a = 0$) [51]. The resulting nonconvex shrinkage-induced penalties can be regarded as the close approximation of the l_0 and low-rank penalties, which are to have the simple generalized shrinkage mappings for solving it.

Remark 1: While completing this article, we become aware of [55] and [56] using similar shrinkage mappings for the NMF-based blind unmixing model. Note that a generalized shrinkage mapping was utilized in the L_p-NMF unmixing model for sparse coding [55]. One major difference between [55] and our work is that the generalized shrinkage mapping in [55] is employed to solve the explicit L_p quasi-norms regularized sparse minimization problem, while, in our case, a new implicitly defined penalty function whose proximal mapping is the specified generalized shrinkage mapping has

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been introduced. The firm thresholding mapping was also utilized in the generalized minimax concave (GMC)-NMF unmixing model [56]. The penalty function whose proximal mapping is the firm thresholding mapping used in this work can be regarded as a special case of the GMC penalty. Furthermore, the shrinkage mappings used in [55] and [56] only approximate the l_0 regularizer, whereas our developed is a further extension version that can approximate both the l_0 and low-rank regularizers.

B. HSUNSP

As introduced in Section III-A, the proposed HSUNSP framework utilizes the nonconvex shrinkage-induced penalties to approximate model (8), which can be uniformly expressed as

$$\begin{aligned} \min_{\mathbf{X}} \quad & \frac{1}{2}\|\mathbf{AX}-\mathbf{Y}\|_F^2+\lambda\ G_1(\mathbf{X})+\tau\ G_2(\sigma(\mathbf{X})) \\ \text{s.t.} \quad & \mathbf{X}\geq 0 \end{aligned} \tag{19}$$

where $\sigma(\mathbf{X})$ denotes the vector of singular values of $\mathbf{X}$. Both G_1 and G_2 can be the firm thresholding-induced penalty function or the generalized shrinkage-induced penalty functions. In fact, (19) contains four different manners to approximate (8). When $G_1=G_2=G_f$ or $G_1=G_2=G_s$, (19) is an approximation of (8) with $c=0$, and we term it hyperspectral sparse unmixing via firm thresholding mapping (HSUFTM) or hyperspectral sparse unmixing via generalized shrinkage mapping (HSUGSM); when $G_1=\|\cdot\|_{\mathbf{w}_j,2,1}$ and $G_2=G_f$, or $G_1=g$ and $G_2=G_s$, (19) is an approximation of (8) with $c=\{2,0\}$, and we term it hyperspectral collaborative sparse unmixing via firm thresholding mapping (HCSUFTM) or hyperspectral collaborative sparse unmixing via generalized shrinkage mapping (HCSUGSM).

The subdifferential of the polyhedral convex function appearing quite often in the proof of Theorem 1 is the indicator function of the nonnegative orthant, as described in Lemma 1.

Lemma 1: Let $\mathbf{x}\in\mathbb{R}^m$, and the subgradients $\mathbf{x}^*$ of $\delta_{\mathbb{R}_+^m}$ at $\mathbf{x}$ form the normal cone to the nonnegative orthant at $\mathbf{x}$, so

$$\partial\delta_{\mathbb{R}_+^m}(\mathbf{x})=\left\{\mathbf{x}^*=\left(\mathbf{x}_1^*,\ldots,\mathbf{x}_m^*\right)|\mathbf{x}^*\leq 0,\left\langle\mathbf{x},\quad\mathbf{x}^*\right\rangle=0\right\}. \tag{20}$$

In other words, $\mathbf{x}^*\in\partial\delta_{\mathbb{R}_+^m}(\mathbf{x})$ is equivalent to m complementary slackness conditions

$$\mathbf{x}_j\geq 0,\ \mathbf{x}_j^*\leq 0,\quad \mathbf{x}_j\mathbf{x}_j^*=0,\quad j=1,\ldots,m. \tag{21}$$

The following theorems are the core ingredient in solving (19).

Theorem 1: Assume that G_1 is one of the following functions.

1) $G_1=G_f$.
2) $G_1=G_s$.
3) $G_1=g$.
4) $G_1=\|\cdot\|_{\mathbf{w}_j,2,1}$.

For any given matrix $\mathbf{C}\in\mathbb{R}^{m\times n}$, $a\in\mathbb{R}$, $\lambda>0$, and $\mu>\lambda$, a local minimizer of the problem

$$\min_{\mathbf{X}}\ \frac{1}{2}\|\mathbf{X}-\mathbf{C}\|_F^2+\lambda\ G_1(\mathbf{X})+\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X}) \tag{22}$$

is given as follows.

1) $\mathbf{X}=S_{\text{firm},\lambda,\mu}(\max\{\mathbf{C},0\})$.
2) $\mathbf{X}=S_{a,\lambda}(\max\{\mathbf{C},0\})$.
3) $\mathbf{X}=S_{a,\lambda}^g(\max\{\mathbf{C},0\})$.
4) $\mathbf{X}=S_{1,\lambda\mathbf{w}_j}^g(\max\{\mathbf{C},0\})$.

Proof: See the Appendix. □

Theorem 2: Assume that G_2 is one of the following functions.

1) $G_2=G_f$.
2) $G_2=G_s$.

For any given matrix $\mathbf{C}\in\mathbb{R}^{m\times n}$, $a\in\mathbb{R}$, $\tau>0$, and $\mu>\tau$, a local minimizer of the problem

$$\min_{\mathbf{X}}\ \frac{1}{2}\|\mathbf{X}-\mathbf{C}\|_F^2+\tau\ G_2(\sigma(\mathbf{X})) \tag{23}$$

is given as follows.

1) $\mathbf{X}=\mathbf{U}_1S_{\text{firm},\tau,\mu}(\Sigma)\mathbf{U}_2^T$.
2) $\mathbf{X}=\mathbf{U}_1S_{a,\tau}(\Sigma)\mathbf{U}_2^T$, where $\mathbf{U}_1\Sigma\mathbf{U}_2^T=\mathbf{C}$ is the singular value decomposition of $\mathbf{C}$.

Proof: The case of 2) was proven in [57], and the proof of case 1) is quite similar to (2) since the same properties of G_f and G_s hold as shown in Proposition 1. □

Now, we employ ADMM to solve (19). An equivalent form of (19) can be written as

$$\begin{aligned} \min_{\mathbf{X}} \quad & \frac{1}{2}\|\mathbf{AX}-\mathbf{Y}\|_F^2 \\ & +\lambda\ G_1(\mathbf{X})+\tau\ G_2(\sigma(\mathbf{X}))+\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X}) \end{aligned} \tag{24}$$

where $\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X})$ is the indicator function, i.e., $\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X})=0$ if $\mathbf{X}>0$ and $\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X})=\infty$ if $\mathbf{X}<0$.

Let $p(\mathbf{X})=\lambda\ G_1(\mathbf{X})+\delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X})$ and $q(\mathbf{X})=\tau\ G_2(\sigma(\mathbf{X}))$, (24) can be written in a simple form

$$\min_{\mathbf{X}}\ \frac{1}{2}\|\mathbf{AX}-\mathbf{Y}\|_F^2+p(\mathbf{X})+q(\mathbf{X}). \tag{25}$$

By introducing two slack variables $\mathbf{V}_1\in\mathbb{R}^{m\times n}$ and $\mathbf{V}_2\in\mathbb{R}^{m\times n}$, (25) can be rewritten as

$$\begin{aligned} \min_{\mathbf{X},\mathbf{V}_1,\mathbf{V}_2} \quad & \frac{1}{2}\|\mathbf{AX}-\mathbf{Y}\|_F^2+p(\mathbf{V}_1)+q(\mathbf{V}_2) \\ \text{s.t.} \quad & \mathbf{X}=\mathbf{V}_1,\ \mathbf{X}=\mathbf{V}_2. \end{aligned} \tag{26}$$

Let $\rho>0$ be a penalty parameter; the augmented Lagrangian function of (26) is defined by

$$\begin{aligned} &L_\rho(\mathbf{X},\mathbf{V}_1,\mathbf{V}_2;\mathbf{D}_1,\mathbf{D}_2) \\ &=\frac{1}{2}\|\mathbf{AX}-\mathbf{Y}\|_F^2+p(\mathbf{V}_1)+q(\mathbf{V}_2) \\ &\quad+\frac{\rho}{2}\|\mathbf{X}-\mathbf{V}_1-\mathbf{D}_1\|_F^2+\frac{\rho}{2}\|\mathbf{X}-\mathbf{V}_2-\mathbf{D}_2\|_F^2-\frac{1}{2\rho}\|\mathbf{D}_1\|_F^2 \\ &\quad-\frac{1}{2\rho}\|\mathbf{D}_2\|_F^2 \end{aligned} \tag{27}$$

where $\mathbf{D}_1\in\mathbb{R}^{m\times n}$ and $\mathbf{D}_2\in\mathbb{R}^{m\times n}$ are dual variables.

We process alternating minimization to the augmented Lagrangian function with respect to each block, as shown in Algorithm 1.

The details of Step 1 in Algorithm 1 are addressed as follows. The minimization of (27) with respect to $\mathbf{X}$ is performed as

$$\mathbf{X}^{k+1}=\operatorname*{argmin}_{\mathbf{X}} L_\rho\left(\mathbf{X},\mathbf{V}_1^k,\mathbf{V}_2^k;\mathbf{D}_1^k,\mathbf{D}_2^k\right)$$

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Algorithm 1 ADMM

Require: Select an initial point $(\mathbf{X}^0, \mathbf{V}_1^0, \mathbf{V}_2^0; \mathbf{D}_1^0, \mathbf{D}_2^0)$. Set $k = 0$, choose $\rho > 0$, $\xi \in (0, \frac{1+\sqrt{5}}{2})$, and $\mu > \lambda, \tau$. Iterate the following steps until the stopping criterion is satisfied:

Step 1. Compute:

$$\mathbf{X}^{k+1} \leftarrow \underset{\mathbf{X}}{\operatorname{argmin}} L_\rho\left(\mathbf{X}, \mathbf{V}_1^k, \mathbf{V}_2^k; \mathbf{D}_1^k, \mathbf{D}_2^k\right),$$
$$\left(\mathbf{V}_1^{k+1}, \mathbf{V}_2^{k+1}\right) \leftarrow \underset{\mathbf{V}_1, \mathbf{V}_2}{\operatorname{argmin}} L_\rho\left(\mathbf{X}^{k+1}, \mathbf{V}_1, \mathbf{V}_2; \mathbf{D}_1^k, \mathbf{D}_2^k\right).$$

Step 2. Update:

$$\mathbf{D}_1^{k+1} \leftarrow \mathbf{D}_1^k - \rho\xi\left(\mathbf{X}^{k+1} - \mathbf{V}_1^{k+1}\right),$$
$$\mathbf{D}_2^{k+1} \leftarrow \mathbf{D}_2^k - \rho\xi\left(\mathbf{X}^{k+1} - \mathbf{V}_2^{k+1}\right),$$
$$k \leftarrow k+1.$$

$$= \underset{\mathbf{X}}{\operatorname{argmin}} \ \frac{1}{2}\|\mathbf{AX} - \mathbf{Y}\|_F^2 + \frac{\rho}{2}\left\|\mathbf{X} - \mathbf{V}_1^k - \mathbf{D}_1^k\right\|_F^2 + \frac{\rho}{2}\left\|\mathbf{X} - \mathbf{V}_2^k - \mathbf{D}_2^k\right\|_F^2. \tag{28}$$

The solution of (28) can be given by

$$\mathbf{X}^{k+1} = \left(\mathbf{A}^T\mathbf{A} + 2\rho\mathbf{I}\right)^{-1}\left[\mathbf{A}^T\mathbf{Y} + \rho\left(\mathbf{V}_1^k + \mathbf{D}_1^k + \mathbf{V}_2^k + \mathbf{D}_2^k\right)\right]. \tag{29}$$

The minimization of (27) with respect to $\mathbf{V}_1$ and $\mathbf{V}_2$ is performed as

$$\begin{aligned}\left(\mathbf{V}_1^{k+1}, \mathbf{V}_2^{k+1}\right) &= \underset{\mathbf{V}_1, \mathbf{V}_2}{\operatorname{argmin}} L_\rho\left(\mathbf{X}^{k+1}, \mathbf{V}_1, \mathbf{V}_2; \mathbf{D}_1^k, \mathbf{D}_2^k\right) \\ &= \underset{\mathbf{V}_1, \mathbf{V}_2}{\operatorname{argmin}} p(\mathbf{V}_1) + q(\mathbf{V}_2) \\ &\quad + \frac{\rho}{2}\left\|\mathbf{X}^{k+1} - \mathbf{V}_1 - \mathbf{D}_1^k\right\|_F^2 + \frac{\rho}{2}\left\|\mathbf{X}^{k+1} - \mathbf{V}_2 - \mathbf{D}_2^k\right\|_F^2.\end{aligned} \tag{30}$$

Since $\mathbf{V}_1$ and $\mathbf{V}_2$ are independent variables, we can handle them separately, that is,

$$\mathbf{V}_1^{k+1} = \underset{\mathbf{V}_1}{\operatorname{argmin}} p(\mathbf{V}_1) + \frac{\rho}{2}\left\|\mathbf{X}^{k+1} - \mathbf{V}_1 - \mathbf{D}_1^k\right\|_F^2 \tag{31}$$

$$\mathbf{V}_2^{k+1} = \underset{\mathbf{V}_2}{\operatorname{argmin}} q(\mathbf{V}_2) + \frac{\rho}{2}\left\|\mathbf{X}^{k+1} - \mathbf{V}_2 - \mathbf{D}_2^k\right\|_F^2. \tag{32}$$

From Theorems 1 and 2, the solutions of (31) and (32) are given in different cases.

1) When $G_1 = G_f$, we have
$$\mathbf{V}_1^{k+1} = S_{\text{firm},\lambda/\rho,\mu}\left(\max\left\{\mathbf{X}^{k+1} - \mathbf{D}_1^k, 0\right\}\right). \tag{33}$$
2) When $G_1 = G_s$, we have
$$\mathbf{V}_1^{k+1} = S_{a,\lambda/\rho}\left(\max\left\{\mathbf{X}^{k+1} - \mathbf{D}_1^k, 0\right\}\right). \tag{34}$$
3) When $G_1 = g$, we have
$$\mathbf{V}_1^{k+1} = S_{a,\lambda/\rho}^g\left(\max\left\{\mathbf{X}^{k+1} - \mathbf{D}_1^k, 0\right\}\right). \tag{35}$$
4) When $G_1 = \|\cdot\|_{\mathbf{w}_j,2,1}$, we have
$$\mathbf{V}_1^{k+1} = S_{1,\lambda\mathbf{w}_j/\rho}^g\left(\max\left\{\mathbf{X}^{k+1} - \mathbf{D}_1^k, 0\right\}\right). \tag{36}$$
5) When $G_2 = G_f$, we have
$$\mathbf{V}_2^{k+1} = \mathbf{U}_1 S_{\text{firm},\tau/\rho,\mu}(\Sigma)\mathbf{U}_2^T \tag{37}$$
where $\mathbf{U}_1\Sigma\mathbf{U}_2^T = \mathbf{X}^{k+1} - \mathbf{D}_2^k$.
6) When $G_2 = G_s$, we have
$$\mathbf{V}_2^{k+1} = \mathbf{U}_1 S_{a,\tau/\rho}(\Sigma)\mathbf{U}_2^T. \tag{38}$$

C. Computational Cost and Convergence Analysis

It is obvious that the proposed framework has the same computational cost as ADSpLRU or JSpBLRU, with complexity $O(m^2n)$ in each iteration. Concerning the convergence, the global convergence and linear convergence rates of the ADMM for the convex problem with two blocks have been well studied in [58] and [59]. Moreover, the convergence of the ADMM for the nonconvex problem has also been studied in [60]. Unfortunately, in our case, the optimization problem (26) does divide into two blocks to handle, but the objective function is nonconvex. The unexplicit form of the nonconvex shrinkage-induced penalties becomes a curse, making it quite arduous to verify the properties in [60, Th. 4]. Some convergence results for the proposed methods on the synthetic data are experimentally visualized in Section IV.

IV. Experiments

To evaluate the proposed framework, several synthetic data with known ground truths and one real data with unknown ground truth are employed in this section. Results of the proposed framework, including HSUFTM, HSUGSM, HCSUFTM, and HCSUGSM, are exhibited. Results of some classic methods, such as SUnSAL [29], CLSUnSAL [30], ADSpLRU [44], and JSpBLRU [45], are also exhibited for comparison purposes. For a fair comparison, the parameters of all the compared methods are tuned to obtain the best unmixing performance. We terminate the proposed framework if the stopping criterion

$$\frac{\left\|\mathbf{X}^{k+1} - \mathbf{X}^k\right\|_F}{\left\|\mathbf{X}^{k+1}\right\|_F} < \text{tol} \tag{39}$$

is satisfied or the maximum number of iterations is achieved. In our experiments, the predefined error tolerance tol is empirically set to 10^{-4}, and the maximum number of iterations is set to 1000. The step size ξ is set to 1.618 [58]. All experiments are finished by using MATLAB R2017b on a MacBook Pro laptop with 2.7-GHz Intel Core i5 and 8-GB memory.

A. Synthetic Data Experiments

In this section, we perform three synthetic datasets to evaluate the unmixing performance of the proposed framework. The spectral library $\mathbf{A}_1$ with 224 bands and 240 endmembers is selected from Chapter 1 of the United States Geological Survey (USGS) digital library.[1] The reflectance values are uniformly distributed in the intervals 0.4 and 2.5 μm. Three synthetic datasets containing 500 pixels are created by randomly selecting a different number of endmembers from $\mathbf{A}_1$ as

[1] Available online: http://speclab.cr.usgs.gov/spectral.lib06

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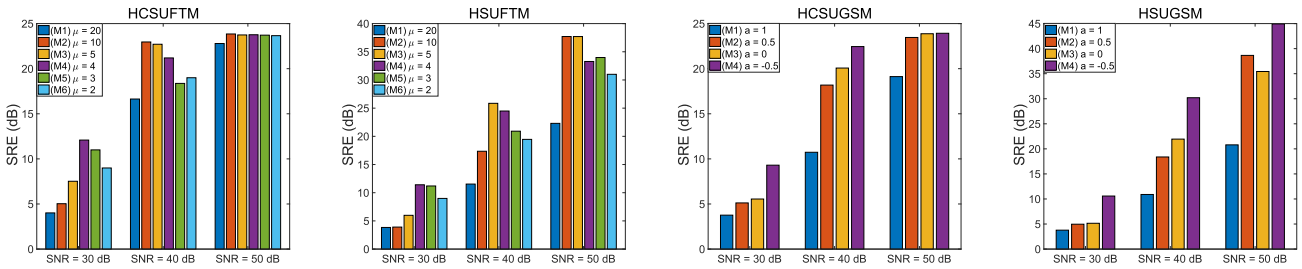


Fig. 2. SREs (dB) values obtained by all the proposed methods using different μ or a values for the DC1 under different SNR levels. (a) HCSUFTM. (b)HSUFTM. (c) HCSUGSM. (d) HSUGSM.

active endmembers: $e_1 = 2$ (denoted by DC1), $e_2 = 4$ (DC2), and $e_3 = 6$ (DC3). For each pixel, the abundances are generated following the Dirichlet distribution and satisfy the abundance nonnegativity constraint and abundance sum-to-one constraint [30]. Then, the Gaussian noise with a different signal-to-noise ratio (SNR $= \mathbf{E}[\|\mathbf{A}\mathbf{x}_i\|_2^2]/\mathbf{E}[\|\mathbf{n}_i\|_2^2]$) of 30, 40, and 50 dB was added to all the synthetic datasets, where $\mathbf{E}[\cdot]$ is the expectation operator.

To quantitatively assess the performance of the different unmixing models, we employ the signal reconstruction error (SRE), the root mean square error (RMSE), and the probability of success (p_s) as the objective metrics, which are defined as

$$\text{SRE(dB)} = 10\log_{10}\left(\frac{\frac{1}{n}\sum_{i=1}^{n}\|\hat{\mathbf{x}}_i\|_2^2}{\frac{1}{n}\sum_{i=1}^{n}\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2}\right) \tag{40}$$

$$\text{RMSE} = \sqrt{\frac{1}{mn}\sum_{i=1}^{n}\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2} \tag{41}$$

$$p_s = P\left(\frac{\|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2^2}{\|\mathbf{x}_i\|_2^2} \le \text{threshold}\right) \tag{42}$$

where $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$ denote the true abundances and the estimated abundances. The unmixing methods are considered successful when $(\|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2^2/\|\mathbf{x}_i\|_2^2) \le 0.316(5\text{ dB})$. In general, the large SRE and p_s or the smaller RMSE indicate the better estimated unmixing performance.

1) Parameter Analysis: As the performance of the proposed framework is sensitive to the parameters μ and a, we first show the SREs (dB) values obtained by all the proposed methods using different μ or a values for the DC1 under different SNR levels in Fig. 2. The firm thresholding mapping-based methods test the unmixing performance of parameter μ ranging from $\{20, 10, 5, 4, 3, 2\}$, as shown in Fig. 2(a) and (b). One can observe that the higher SRE values are obtained with $\mu = 3$ or 4 in the case of SNR $= 30$ dB, while the higher SRE values are obtained with $\mu = 5$ in the case of the higher SNR levels. The value of μ is too large or too small may lead to poor performance. The generalized shrinkage mapping-based methods test the unmixing performance of parameter a ranging from $\{1, 0.5, 0, -0.5\}$, as shown in Fig. 2(c) and (d). We should mention that the HCSUGSM and HSUGSM degenerate to convex models when $a = 1$. It can be clearly observed that the convex cases get lower SREs values than others under all SNR cases. With the decrease in a, the unmixing performance is gradually improved until $a = -0.5$ achieves the best quantitative results. Thus, we fix $\mu = 3$ and $a = -0.5$ in the following experiments.

We further assess the effect of the regularization parameters λ and τ of the proposed methods on the unmixing performance. For this purpose, the regularization parameters λ and τ are tested from the following finite set $\{10, 5, 1, 0.5, 0.1, 0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001, 0.00001, 0.000001\}$. Fig. 3 plots the SREs values as a function of the two regularization parameters for the DC1 under different SNR values. One can observe that all proposed methods achieve low SRE values when either λ or τ is too large. The curves of the same methods with different SNR levels exhibit similar behaviors. The SRE values increase as the parameters λ or τ decrease, and the SRE values are stable when the parameters are relatively small. This phenomenon demonstrates the positive influence of the proper selection of the two parameters on the unmixing performance. Moreover, one can observe that the highest SRE values are achieved for all the methods when the optimal τ is much larger than the optimal λ, demonstrating the significance of the low-rank regularizer on the unmixing model. Furthermore, one can observe that the optimal parameters are different under different noise levels. A general trend is that the highest SRE values are obtained with smaller values of the λ and τ as SNR increases. These observations actually give us a lighthouse for hyperparameter selection. The smaller hyperparameters are selected for unmixing in the case of a higher SNR level; otherwise, the larger hyperparameters are selected.

2) Performance Comparisons: Tables I–III list the quantitative results for three different metrics (SRE, RMSE, and p_s), the running time for each iteration, and the regularization parameters for each method on the synthetic datacubes. Ten experiments were performed, and the average value was taken as the result of each method. To fairly evaluate the unmixing performance, we set the optimal regularization parameters for all the tested methods. We first compare the proposed methods with state-of-the-art methods. From Tables I to III, one can observe that the SUnSAL and CLSUnSAL yield poor quantitative results in all the cases since only the spectral information is considered, while, for the methods considering both spectral information and spatial information, our proposed methods achieve better quantitative results in all the cases. For example, for the case of DC1 with

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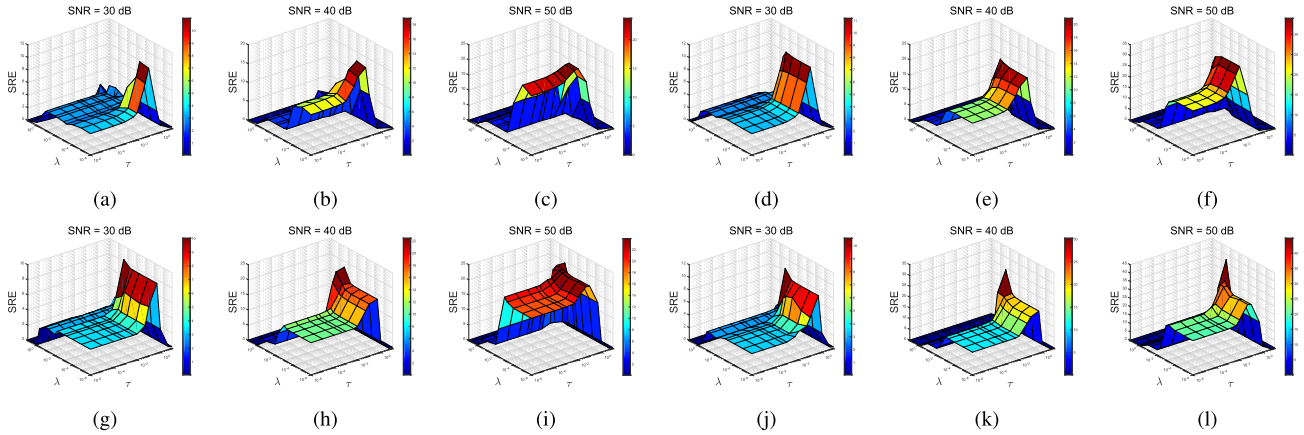


Fig. 3. SREs (dB) values as a function of the regularization parameters λ and τ for the DC1 under different SNR levels. (a) HCSUFTM (30 dB). (b) HCSUFTM (40 dB). (c) HCSUFTM (50 dB). (d) HSUFTM (30 dB). (e) HSUFTM (40 dB). (f) HSUFTM (50 dB). (g) HCSUGSM (30 dB). (h) HCSUGSM (40 dB). (i) HCSUGSM (50 dB). (j) HSUGSM (30 dB). (k) HSUGSM (40 dB). (l) HSUGSM (50 dB).

TABLE I

QUANTITATIVE RESULTS OF THE NUMERICAL EXPERIMENTS ON THE DC1. THE BEST TWO ARE MARKED IN BOLD

SNR	Parameters	SUnSAL	CLSUnSAL	ADSpLRU	JSpBLRU	HCSUFTM	HCSUGSM	HSUFTM	HSUGSM
30 dB	SRE (dB)	3.3614	3.7881	5.8404	6.5109	**10.9976**	10.0196	**11.1943**	10.5855
	RMSE	0.0356	0.0339	0.0268	0.0248	**0.0148**	0.0165	**0.0144**	0.0155
	p_s	0.4060	0.4120	0.6280	0.7200	0.9660	**0.9960**	0.9700	**0.9980**
	time ($\times 10^{-3}$ s)	**2.6**	**3.1**	24.5	23.0	20.9	27.6	22.6	22.5
	λ τ	λ = 0.0001	λ = 0.001	λ = 0.000001 τ = 0.1	λ = 0.00001 τ = 0.05	λ = 0.00001 τ = 0.1	λ = 0.005 τ = 0.5	λ = 0.0005 τ = 0.1	λ = 0.005 τ = 0.5
40 dB	SRE (dB)	10.4231	10.9174	19.2578	**23.6400**	18.8568	22.4517	20.9132	**30.2038**
	RMSE	0.0158	0.0149	0.0057	**0.0034**	0.0060	0.0040	0.0047	**0.0016**
	p_s	0.9740	0.9840	**1.0000**	0.9960	0.9960	0.9960	**1.0000**	**1.0000**
	time ($\times 10^{-3}$ s)	**2.7**	**3.1**	24.5	22.9	21.0	22.8	20.5	22.1
	λ τ	λ = 0.00001	λ = 0.0005	λ = 0.0001 τ = 0.1	λ = 0.0001 τ = 0.05	λ = 0.000001 τ = 0.05	λ = 0.005 τ = 0.5	λ = 0.001 τ = 0.05	λ = 0.005 τ = 0.5
50 dB	SRE (dB)	19.7796	20.5591	33.0200	23.9523	23.7176	23.9265	**33.9820**	**44.9363**
	RMSE	0.0054	0.0049	0.0012	0.0033	0.0034	0.0033	**0.0010**	**0.0003**
	p_s	**1.0000**	**1.0000**	**1.0000**	0.9960	0.9960	0.9960	**1.0000**	**1.0000**
	time ($\times 10^{-3}$ s)	**2.7**	**2.9**	24.0	22.3	21.7	22.8	20.3	21.6
	λ τ	λ = 0.00001	λ = 0.0005	λ = 0.00001 τ = 0.05	λ = 0.0001 τ = 0.05	λ = 0.00001 τ = 0.01	λ = 0.005 τ = 0.1	λ = 0.000001 τ = 0.01	λ = 0.005 τ = 0.5

TABLE II

QUANTITATIVE RESULTS OF THE NUMERICAL EXPERIMENTS ON THE DC2. THE BEST TWO ARE MARKED IN BOLD

SNR	Parameters	SUnSAL	CLSUnSAL	ADSpLRU	JSpBLRU	HCSUFTM	HCSUGSM	HSUFTM	HSUGSM
30 dB	SRE (dB)	3.6205	4.6752	5.1441	5.6079	5.5029	**5.6605**	5.3071	**5.6142**
	RMSE	0.0267	0.0236	0.0224	0.0213	0.0215	**0.0211**	0.0220	**0.0212**
	p_s	0.4260	0.6080	0.6340	0.6540	0.6520	**0.6640**	0.6460	**0.6580**
	time ($\times 10^{-3}$ s)	**2.6**	**3.9**	24.5	22.5	21.0	22.9	20.4	22.3
	λ τ	λ = 0.005	λ = 0.1	λ = 0.001 τ = 0.5	λ = 0.001 τ = 1	λ = 0.001 τ = 0.5	λ = 0.05 τ = 1	λ = 0.05 τ = 0.5	λ = 0.05 τ = 1
40 dB	SRE (dB)	7.4774	8.4451	12.3592	8.5458	13.3891	**14.9362**	12.4968	**15.7351**
	RMSE	0.0171	0.0153	0.0098	0.0151	0.0087	**0.0073**	0.0096	**0.0066**
	p_s	0.7960	0.8840	**0.9980**	0.8360	0.9820	0.9960	0.9960	**1.0000**
	time ($\times 10^{-3}$ s)	**2.6**	**3.0**	27.5	22.8	20.7	23.4	20.2	22.6
	λ τ	λ = 0.0001	λ = 0.001	λ = 0.00001 τ = 0.01	λ = 0.00001 τ = 0.01	λ = 0.00001 τ = 0.01	λ = 0.001 τ = 0.1	λ = 0.000001 τ = 0.01	λ = 0.001 τ = 0.1
50 dB	SRE (dB)	15.6341	16.3400	**28.2038**	18.2441	24.0338	25.1665	24.2671	**35.8367**
	RMSE	0.0067	0.0062	**0.0016**	0.0050	0.0025	0.0022	0.0025	**0.0007**
	p_s	**1.0000**	**1.0000**	**1.0000**	0.9960	0.9960	0.9960	**1.0000**	**1.0000**
	time ($\times 10^{-3}$ s)	**2.8**	**3.1**	24.3	22.8	21.0	22.4	20.1	22.1
	λ τ	λ = 0.00001	λ = 0.0001	λ = 0.00001 τ = 0.01	λ = 0.00001 τ = 0.001	λ = 0.00001 τ = 0.01	λ = 0.001 τ = 0.1	λ = 0.000001 τ = 0.005	λ = 0.001 τ = 0.1

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TABLE III

QUANTITATIVE RESULTS OF THE NUMERICAL EXPERIMENTS ON THE DC3. THE BEST TWO ARE MARKED IN BOLD

SNR	Parameters	SUnSAL	CLSUnSAL	ADSpLRU	JSpBLRU	HCSUFTM	HCSUGSM	HSUFTM	HSUGSM
30 dB	SRE (dB)	2.4860	3.8563	5.0832	**7.1892**	**6.7306**	5.9002	5.2125	6.6176
	RMSE	0.0258	0.0220	0.0191	**0.0150**	**0.0158**	0.0174	0.0189	0.0160
	p_s	0.2140	0.3660	0.5880	**0.7740**	0.7560	0.6880	0.6100	**0.7680**
	time ($\times 10^{-3}$ s)	**3.0**	**5.4**	24.6	22.8	21.2	23.1	20.5	25.6
	λ τ	λ = 0.01	λ = 0.1	λ = 0.001 τ = 0.1	λ = 0.0005 τ = 0.1	λ = 0.001 τ = 0.1	λ = 0.05 τ = 1	λ = 0.01 τ = 0.1	λ = 0.01 τ = 0.5
40 dB	SRE (dB)	5.3274	6.1948	7.4263	7.3846	7.5018	**10.0744**	7.6452	**9.7196**
	RMSE	0.0186	0.0168	0.0146	0.0147	0.0145	**0.0108**	0.0143	**0.0112**
	p_s	0.6220	0.7360	0.7880	0.7820	0.7940	**0.9920**	0.8020	**0.9940**
	time ($\times 10^{-3}$ s)	**2.6**	**4.3**	25.9	23.1	20.8	24.2	21.2	22.3
	λ τ	λ = 0.001	λ = 0.05	λ = 0.0001 τ = 0.05	λ = 0.0005 τ = 0.01	λ = 0.000001 τ = 0.05	λ = 0.001 τ = 0.05	λ = 0.000001 τ = 0.05	λ = 0.000001 τ = 0.05
50 dB	SRE (dB)	11.1480	11.5234	22.4923	11.5204	18.9661	**22.6079**	18.8108	**27.0086**
	RMSE	0.0095	0.0091	0.0026	0.0091	0.0039	**0.0025**	0.0039	**0.0015**
	p_s	0.9980	0.9980	**1.0000**	0.9820	0.9960	0.9960	**1.0000**	**1.0000**
	time ($\times 10^{-3}$ s)	**2.8**	**3.1**	24.5	22.6	20.6	26.4	20.7	23.1
	λ τ	λ = 0.00001	λ = 0.0001	λ = 0.000001 τ = 0.005	λ = 0.000001 τ = 0.001	λ = 0.000001 τ = 0.005	λ = 0.001 τ = 0.05	λ = 0.00001 τ = 0.005	λ = 0.001 τ = 0.05

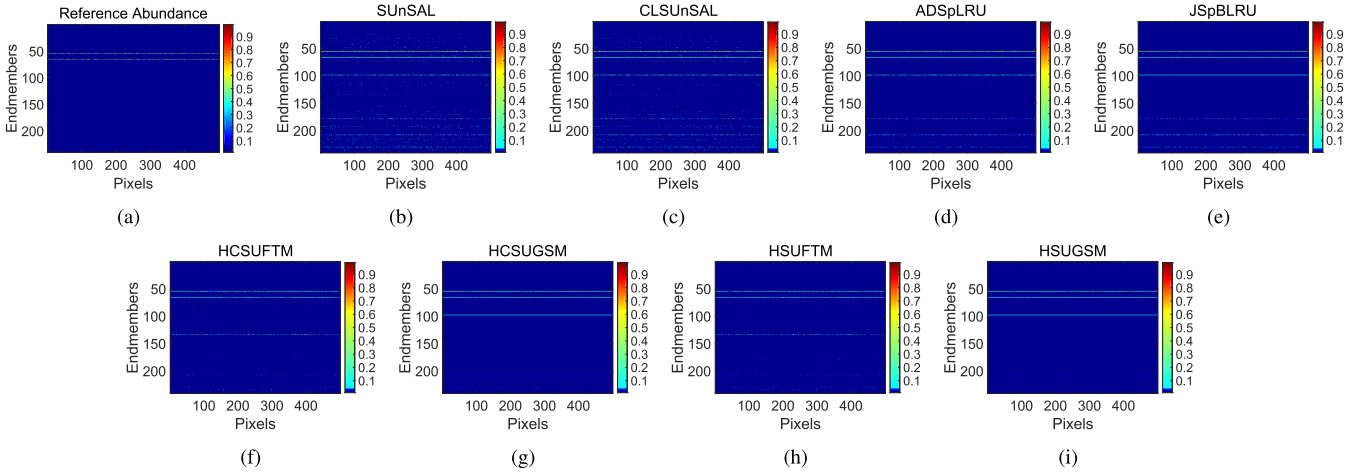


Fig. 4. Reference abundance and the difference maps between the true abundances and the estimated abundances using different unmixing methods for the DC1 with SNR = 30 dB. (a) Reference abundance. (b) SUnSAL. (c) CLSUnSAL. (d) ADSpLRU. (e) JSpBLRU. (f) HCSUFTM. (g) HCSUGSM. (h) HSUFTM. (i) HSUGSM.

SNR = 30 dB, compared with ADSpLRU or JSpBLRU, the SRE values obtained by the proposed method are increased by about 4 dB, the RMSE values are decreased by about 0.008, and the p_s values are increased by more than 20%. This provides experimental evidence of the potential of the nonconvex shrinkage penalties in solving the hyperspectral sparse unmixing problem. Moreover, we also compare the proposed firm thresholding mapping-based methods and generalized shrinkage mapping-based methods. Again from Tables I to III, one can observe that the quantitative results obtained by the generalized shrinkage mapping-based methods are better than the firm thresholding mapping-based methods. Especially, the SRE values obtained by the HSUGSM increase by about 10 dB compared with HSUFTM under SNR = 50 dB. In terms of the running time for each iteration, the SUnSAL and CLSUnSAL are the fastest, while the methods considering both spectral information and spatial information, such as ADSpLRU, JSpBLRU, and our proposed methods, consume more time. This is consistent with the fact that it takes excessive time to handle the singular value decomposition in each iteration.

Fig. 4 shows the difference maps between the true abundances and the estimated abundances using different unmixing methods for the DC1 with SNR = 30 dB. Visually, Fig. 4(a) and (b) shows many noisy points in many lines due to the lack of incorporation of spatial information. Taking the rest of the algorithms, by and large, those methods considering both spectral information and spatial information show fewer noise points, as shown in Fig. 4(c)–(h). In other words, most of the endmembers are active for the SUnSAL and CLSUnSAL, while just a small number of endmembers are active for the other unmixing methods. This indirectly demonstrates that only a few endmembers are present in each pixel. Moreover, one can observe that the performance of the proposed methods outperforms the ADSpLRU and JSpBLRU, and the superiority

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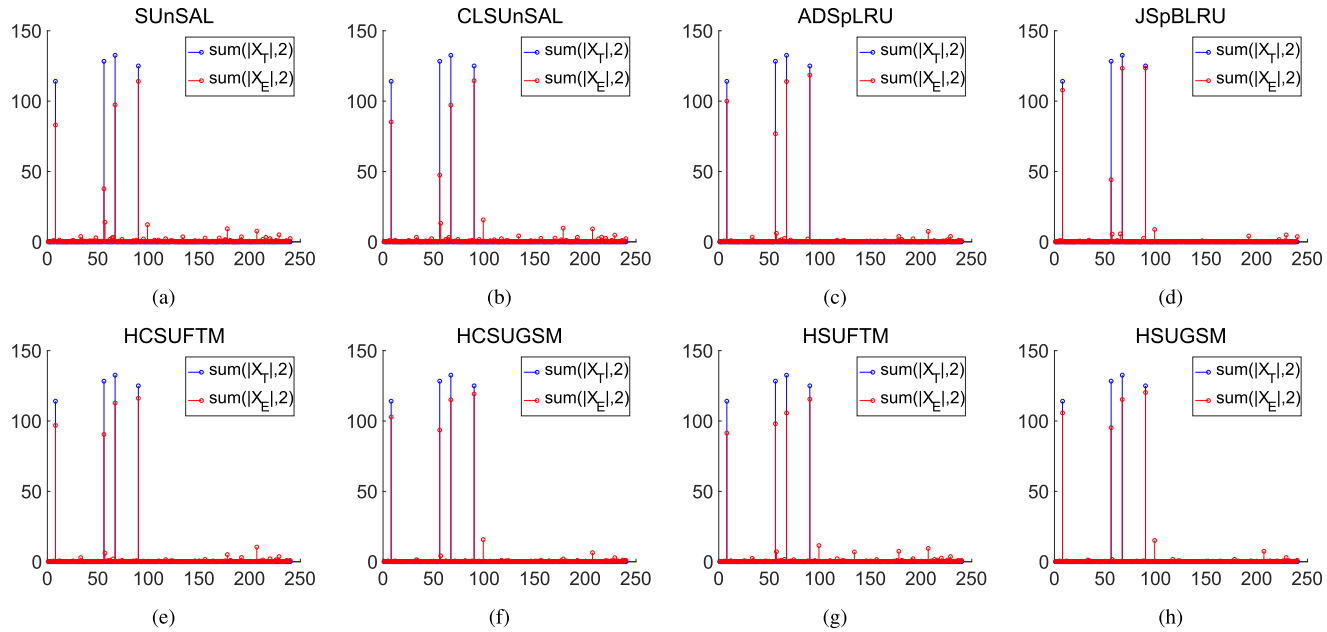


Fig. 5. Sum of all pixels corresponding to each endmember for the estimated abundances $\mathbf{X}_E$ using different unmixing methods for the DC2 with SNR $= 40$ dB. (a) SUnSAL. (b) CLSUnSAL. (c) ADSpLRU. (d) JSpBLRU. (e) HCSUFTM. (f) HCSUGSM. (g) HSUFTM. (h) HSUGSM.

is quite obvious for the HCSUGSM and HSUGSM. These results reveal the superiority of the proposed methods to identify the endmembers in the abundance.

To further verify the effectiveness of the proposed methods, Fig. 5 shows the sum of all pixels corresponding to each endmember for the estimated abundances using different unmixing methods for the DC2 with SNR $= 40$ dB. The blue and red lines in Fig. 5 represent the true abundances $\mathbf{X}_T$ and estimated abundances $\mathbf{X}_E$, respectively. By and large, conformity of the red and blue lines means better unmixing performance. At first glance, all the methods seem to identify the true endmember. If we focus on the second endmember, we can observe that the red lines in Fig. 5(e)–(h) are closer to the blue one than in Fig. 5(a)–(d). As expected, the performance of the proposed methods is superior to that of other methods, indicating the effectiveness and superiority of the nonconvex shrinkage penalties in handling unmixing model. Fig. 6 shows the RMSE values obtained by different unmixing methods under different SNR levels for the DC2. One can clearly observe that all the SSLRR-based unmixing methods achieve better results than the SUnSAL and CLSUnSAL, especially for the low SNR levels. The proposed methods generally provide better RMSE values than the ADSpLRU and JSpBLRU. This is because the low-rank regularization in the SSLRR-based models can suppress the noise and, thus, obtain a more accurate solution.

3) Convergence Comparisons: Convergence analysis of the proposed unmixing methods is experimentally performed on the DC3. Fig. 7(a)–(c) shows the variations of the RMSE values versus the iterations for the proposed unmixing methods under different SNR levels. It can be observed that the curve trend of all proposed methods is considerably similar, while the convergence characteristic of all the methods becomes steady with increasing iterations in all cases. It is clear that it is enough to set the maximum number of iterations to 1000.

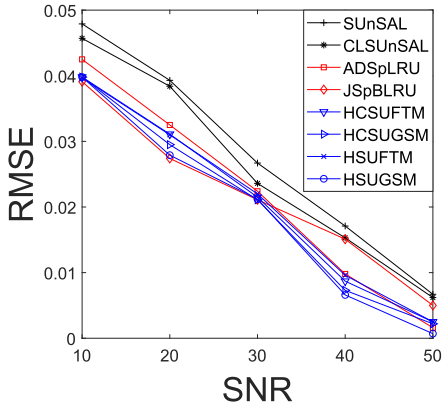


Fig. 6. RMSE values obtained by different unmixing methods under different SNR levels for the DC2.

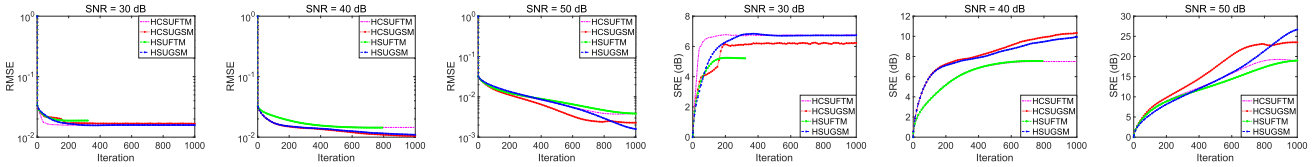
Moreover, Fig. 7(d)–(f) shows the variations of the SRE values versus the iterations for the proposed unmixing methods under different SNR levels. Again, it can be observed that the curves of different methods exhibit similar behaviors, while the SRE peak value for each method can be reached within the maximum number of iterations. Although we cannot theoretically give the convergence of the proposed algorithm, these experiments indirectly demonstrate the robustness of our proposed algorithm.

B. Real Data Experiments

The real dataset in the experiments is the publicly available Airborne Visible Infrared Spectrometer (AVIRIS) Cuprite data,[2] which is composed of various minerals and quite prevalent for hyperspectral analysis. We selected a 250×191 pixels

[2]Available online: http://aviris.jpl.nasa.gov/html/aviris.freedata.html

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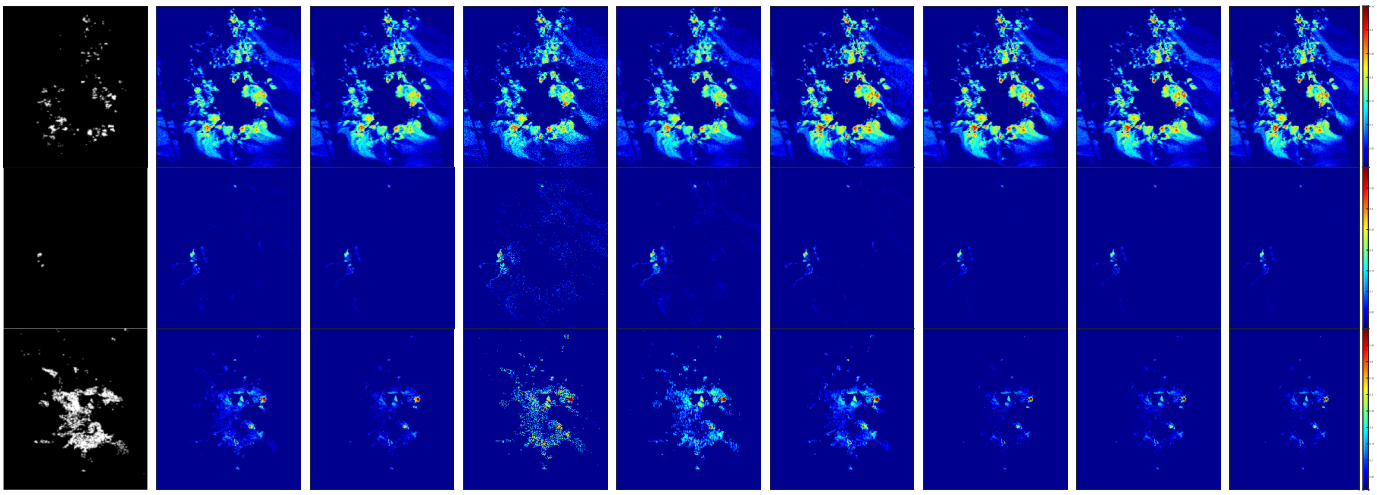
(a) (b) (c) (d) (e) (f)

Fig. 7. Convergence analysis of the proposed unmixing methods is experimentally performed on the DC3 under different SNR levels. (a) Variations of the RMSE values versus the iteration for the proposed methods (SNR = 30 dB). (b) Variations of the RMSE values versus the iteration for the proposed methods (SNR = 40 dB). (c) Variations of the RMSE values versus the iteration for the proposed methods (SNR = 50 dB). (d) Variations of the SRE (dB) values versus the iteration for the proposed methods (SNR = 30 dB). (e) Variations of the SRE (dB) values versus the iteration for the proposed methods (SNR = 40 dB). (f) Variations of the SRE (dB) values versus the iteration for the proposed methods (SNR = 50 dB).

TABLE IV

QUANTITATIVE RESULTS OF THE NUMERICAL EXPERIMENTS ON THE CUPRITE DATA. THE BEST TWO ARE MARKED IN BOLD

Parameters	SUnSAL	CLSUnSAL	ADSpLRU	JSpBLRU	HCSUFTM	HCSUGSM	HSUFTM	HSUGSM
RE	0.0143	**0.0056**	0.1307	0.0665	0.0232	0.0064	0.0063	**0.0062**
time (s)	**1.1**	**1.1**	7.1	7.0	6.7	6.7	6.7	6.9
λ τ	$\lambda = 0.001$	$\lambda = 0.01$	$\lambda = 0.001$ $\tau = 0.001$	$\lambda = 0.001$ $\tau = 0.001$	$\lambda = 0.001$ $\tau = 0.001$	$\lambda = 0.001$ $\tau = 0.001$	$\lambda = 0.001$ $\tau = 0.001$	$\lambda = 0.001$ $\tau = 0.001$



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 8. Reference abundance maps obtained by Tricorder software and the estimated abundance maps obtained by all the different methods for the minerals: alunite (the first row), buddingtonite (the second row), chalcedony (the third row). (a) Reference abundance maps. (b) SUnSAL. (c) CLSUnSAL. (d) ADSpLRU. (e) JSpBLRU. (f) HCSUFTM. (g) HCSUGSM. (h) HSUFTM. (i) HSUGSM.

data subset to evaluate the unmixing performance between the proposed methods and the compared methods. The dataset contains 224 spectral bands ranging from 0.4 to 2.5 μm with 10-nm spectral resolution. Prior to the analysis, we remove the bands 1, 2, 105–115, 150–170, 223, and 224 since the interference of noise and water absorption, leaving a total of 188 effective spectral bands. The USGS spectral library $\mathbf{A}$ used here is to remove the corresponding bands, with 188 bands and 498 endmembers.

Since the ground-truth information of the Cuprite data is unknown, we employ a new measure of reconstruction error (RE) to quantitatively assess the performance of the different unmixing methods from the point of view of data reconstruction, which is defined as

$$\mathrm{RE} = \sqrt{\frac{1}{mn}\sum_{i=1}^{n}\|\mathbf{y}_i - \hat{\mathbf{y}}_i\|_2^2} \tag{43}$$

where $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ denote the real pixel and the reconstructed pixel vectors, respectively. Note that the RE can only roughly assess the abundances estimations [61].

Table IV lists the RE values, the running time for each iteration, and the regularization parameters for each method on the Cuprite data. The regularization parameter settings of the comparison algorithm are consistent with those suggested in the corresponding papers, that is, SUnSAL with $\lambda = 0.001$ as in [29], CLSUnSAL with $\lambda = 0.01$ as in [30], and both ADSpLRU and JSpBLRU with $\lambda = 0.001$ and $\tau = 0.001$ as in [44] and [45]. Since the setting of regularization parameters has a certain influence on the RE values for a certain unmixing model, intuitively, the smaller parameters imply the dominant effect of the fitting term $(1/2)\|\mathbf{AX} - \mathbf{Y}\|_F^2$, which leads to higher RE values. Conversely, the larger the parameters result in lower RE values. To fairly assess the unmixing performance, the regularization parameters used in our proposed methods

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are the same as in ADSpLRU and JSpBLRU. From Table IV, one can observe that the CLSUnSAL can obtain the smallest RE values, while, for the methods considering both spectral and spatial information, our proposed methods can achieve smaller RE values compared with ADSpLRU and JSpBLRU. Again, one can observe that the methods containing one regularizer, such as the SUnSAL and CLSUnSAL, are the fastest for each iteration, and the methods containing two regularizers are slow but still acceptable.

For visual comparison, Fig. 8 shows the estimated abundance maps obtained by all the different methods for the mineral: alunite, buddingtomite, and chalcedony. From Fig. 8, one can observe that all methods produce similar abundance maps. However, the abundance maps estimated by the proposed methods are generally higher in the regions considered as the mineral in comparison with other methods for the Alunite. For the buddingtomite and chalcedony, apparently, the proposed methods suppress the noise and maintain the structural information best. Generally, from a qualitative viewpoint, we can conclude that the proposed methods show comparable unmixing results on the real data.

V. Conclusion

In this article, we exhibited experimental evidence of the potential of the two thresholding mappings in solving the hyperspectral sparse unmixing problem. The proposed framework provides a new manner to approximate the SSLRR-based unmixing models. An important aspect of the proposed framework is that it does not require explicit forms of penalties but only relevant thresholding mappings, which is more flexible for most state-of-the-art algorithms for unmixing. Extensive numerical experiments on synthetic and real hyperspectral data demonstrated the effectiveness of the proposed framework over existing SSLRR-based unmixing models. In further studies, we will adapt the proposed methods to be applicable to the unsupervised-based approaches. Moreover, proving the convergence of the proposed framework is also a promising topic.

Appendix A
Proof of Theorem 1

Proof: We divide the proof into four parts.

First, when $G_1 = G_f$, we want to prove that a local minimizer of the problem

$$\min_{\mathbf{Z}} \frac{1}{2}\|\mathbf{Z}-\mathbf{C}\|_F^2 + \lambda\ G_f(\mathbf{Z}) + \delta_{\mathbb{R}_+^{m\times n}}(\mathbf{Z}) \tag{44}$$

is given by

$$\mathbf{Z} = S_{\text{firm},\lambda,\mu}(\max\{\mathbf{C},0\}). \tag{45}$$

From Proposition 1, we know that $G_f(\mathbf{X}) = \sum_{i,j} g_f(\mathbf{X}_{ij})$. Thus, (44) is element-separable and can be divided into $m \times n$ subproblem.

For the case of $\mathbf{Z}_{ij} \in \mathbb{R}^1$, considering the following problem:

$$\min_{\mathbf{Z}_{ij}} \frac{1}{2}\|\mathbf{Z}_{ij}-\mathbf{C}_{ij}\|_2^2 + \lambda\ g_f\big(\mathbf{Z}_{ij}\big) + \delta_{\mathbb{R}_+^1}\big(\mathbf{Z}_{ij}\big). \tag{46}$$

It suffices to verify that the first-order optimality condition of (46) holds, that is,

$$0 \in \mathbf{Z}_{ij} - \mathbf{C}_{ij} + \lambda\partial g_f\big(\mathbf{Z}_{ij}\big) + \partial\delta_{\mathbb{R}_+^1}\big(\mathbf{Z}_{ij}\big). \tag{47}$$

From the first-order optimality condition of (13), we know that

$$0 \in \mathbf{X}_{ij} - \mathbf{C}_{ij} + \lambda\partial g_f\big(\mathbf{X}_{ij}\big). \tag{48}$$

1) When $\mathbf{C}_{ij} > 0$, which means that

$$\mathbf{Z}_{ij} = \mathbf{X}_{ij}. \tag{49}$$

By lemma 1, we have

$$0 \in \partial\delta_{\mathbb{R}_+^1}\big(\mathbf{Z}_{ij}\big). \tag{50}$$

Combining (48)–(50), (47) holds directly.

2) When $\mathbf{C}_{ij} \le 0$, which means that

$$\mathbf{Z}_{ij} = 0. \tag{51}$$

Thus, it is enough to show that

$$0 \in \lambda\partial g_f(0) + \partial\delta_{\mathbb{R}_+^1}(0) - \mathbf{C}_{ij}. \tag{52}$$

By Proposition 1 and Lemma 1, (52) is equivalent to the existence of $d \in [-1,1]$ and $e \le 0$ such that

$$\mathbf{C}_{ij} = \lambda d + e \tag{53}$$

which is held directly.

Second, when $G_1 = G_s$, the proof of this case is very similar to (1) since the same properties of G_f and G_s hold.

Third, when $G_1 = g$, we want to prove that a local minimizer of the problem

$$\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{X}-\mathbf{C}\|_F^2 + \lambda\ g\big(\|\mathbf{X}\|_{2,1}\big) + \delta_{\mathbb{R}_+^{m\times n}}(\mathbf{X}) \tag{54}$$

is given by

$$\mathbf{X} = T_{a,\lambda}(\max\{\mathbf{C},0\}). \tag{55}$$

Note that (54) is row-separable and can be divided into m subproblem. To simplify the notations, let $\mathbf{Z}^i = \max\{\mathbf{C}^i,0\}$, $\mathbf{X}^i$, $\mathbf{Z}^i$, and $\mathbf{C}^i$ be ith row of $\mathbf{X}$, $\mathbf{Z}$, and $\mathbf{C}$.

For the case of $\mathbf{X}^i \in \mathbb{R}^n$, consider the following problem:

$$\min_{\mathbf{X}^i} \frac{1}{2}\|\mathbf{X}^i-\mathbf{C}^i\|_2^2 + \lambda\ g\big(\|\mathbf{X}^i\|_2\big) + \delta_{\mathbb{R}_+^n}\big(\mathbf{X}^i\big). \tag{56}$$

1) When $\mathbf{X}^i = 0$, which means that

$$\|\mathbf{Z}^i\|_2 \le \lambda. \tag{57}$$

It is enough to show that the first-order optimality condition of (56) holds, that is,

$$0 \in \lambda g'(0)\partial\|\cdot\|_2(0) - \mathbf{C}^i + \partial\delta_{\mathbb{R}_+^n}(0). \tag{58}$$

This is equivalent to find $\|\mathbf{d}\|_2 \le 1$ and $\mathbf{h} \le 0$ such that

$$\mathbf{C}^i = \lambda\mathbf{d} + \mathbf{h}. \tag{59}$$

Note that $\mathbf{Z}^i$ is the minimizer of

$$\min_{\mathbf{Z}^i} \frac{1}{2}\|\mathbf{Z}^i-\mathbf{C}^i\|_2^2 + \delta_{\mathbb{R}_+^n}\big(\mathbf{Z}^i\big). \tag{60}$$

□

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Then, we have

$$0 \in \mathbf{Z}^i - \mathbf{C}^i + \partial\delta_{\mathbb{R}^n_+}(\mathbf{Z}^i). \tag{61}$$

Thus, there exists $\mathbf{h} \leq 0$ such that

$$\mathbf{C}^i = \mathbf{Z}^i + \mathbf{h}. \tag{62}$$

Combining (57), the proof is complete for $\mathbf{X}^i = 0$.

2) When $\mathbf{X}^i \neq 0$. By the definition of $T_{a,\lambda}$, we have that $\mathbf{X}^i = t\mathbf{Z}^i$ for some $t > 0$. It is enough to show that

$$0 \in \mathbf{X}^i - \mathbf{C}^i + \lambda g'(\|\mathbf{X}^i\|_2)\frac{\mathbf{X}^i}{\|\mathbf{X}^i\|_2} + \partial\delta_{\mathbb{R}^n_+}(\mathbf{X}^i). \tag{63}$$

Since $\mathbf{X}^i = T_{a,\lambda}(\mathbf{Z}^i)$, we have

$$\lambda g'(\|\mathbf{X}^i\|_2)\frac{\mathbf{X}^i}{\|\mathbf{X}^i\|_2} + \mathbf{X}^i - \mathbf{Z}^i = 0. \tag{64}$$

By substituting (64) into (63), we only need to show that

$$0 \in \mathbf{Z}^i - \mathbf{C}^i + \partial\delta_{\mathbb{R}^n_+}(\mathbf{X}^i). \tag{65}$$

Construct the following problem:

$$\min_{\mathbf{W}^i} \frac{1}{2}\|\mathbf{W}^i - t\mathbf{C}^i\|_2^2 + t\delta_{\mathbb{R}^n_+}(\mathbf{W}^i). \tag{66}$$

From the first-order condition of (66), we have

$$0 \in \mathbf{W}^i - t\mathbf{C}^i + t\partial\delta_{\mathbb{R}^n_+}(\mathbf{W}^i). \tag{67}$$

Since $\mathbf{W}^i = t\mathbf{Z}^i = \mathbf{X}^i$, (67) is equal to

$$0 \in \mathbf{X}^i - t\mathbf{C}^i + t\partial\delta_{\mathbb{R}^n_+}(\mathbf{X}^i). \tag{68}$$

Dividing through by t completes the proof for $\mathbf{X}^i \neq 0$.

Fourth, when $G_1 = \|\cdot\|_{\mathbf{w}_j,2,1}$, we assume that the weighting factor $\mathbf{w}_j$ is fixed in each iteration. The proof is complete by the equivalent of 2) and 4) in [62, Th. 4].

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